DAF-CL: A Robust Multimodal Emotion Classification Framework with Dynamic Audio-Text Fusion and Contrastive Learning

Abstract

Emotion recognition is a key component of affect-aware human–computer interaction, enabling conversational agents, virtual assistants, and other intelligent systems to respond more naturally to users' emotional states. Although multimodal approaches can improve recognition by exploiting complementary speech, text, and visual information, visual signals are not always available or practical in real-world settings, making speech–text fusion a particularly relevant alternative. The central challenge, however, is not only to extract informative features from each modality, but to determine how heterogeneous acoustic and textual evidence should be combined when their reliability varies across utterances and when emotionally salient acoustic cues are distributed unevenly over time. To address this problem, we propose DAF-CL, a Dynamic Adaptive Fusion with Contrastive Learning framework that performs information fusion at multiple levels. At the acoustic level, overlapping speech chunks are encoded with WavLM and aggregated through a confidence-aware mechanism that combines learned emotional saliency with prediction uncertainty, allowing informative regions to contribute more strongly to the utterance-level audio decision. At the multimodal level, the resulting audio evidence is integrated with RoBERTa-based textual representations through a dynamic late-fusion mechanism whose modality contribution is adapted for each utterance rather than fixed globally. To further improve fusion compatibility, a bidirectional contrastive objective aligns utterance-level acoustic and textual embeddings in a shared latent space, while Structured Supervised Contrastive Learning (SSCL) organizes the acoustic representation space using continuous class-dependent emotion similarities instead of conventional binary inter-class relationships. The complete framework is jointly optimized through classification, cross-modal alignment, and structured contrastive objectives, enabling local acoustic weighting, representation-level alignment, and decision-level modality fusion to operate in a unified architecture. Experimental results show that DAF-CL achieves a Macro F1 score of 87.4% on the evaluation set, improving the corresponding multimodal baseline by 9.0 percentage points. Evaluation on unseen emotion datasets further demonstrates consistent generalization across different recording conditions and data distributions, while ablation studies, design-choice comparisons, representation analysis, and sensitivity experiments verify the contribution of the proposed fusion components. These results indicate that explicitly modeling information reliability, cross-modal compatibility, and adaptive modality contribution provides an effective foundation for more generalizable speech–text emotion recognition.

Keywords: Multimodal emotion recognition; Speech–text fusion; Dynamic adaptive fusion; Cross-modal alignment; Structured supervised contrastive learning; Speech emotion recognition.

1. Introduction

Emotion plays a crucial role in human life, influencing decision-making processes and interpersonal interactions [1]. With the rapid advancement of technology and the increasing integration of artificial intelligence into everyday tasks, there is a growing need to develop AI systems capable of understanding and responding to human emotions effectively [2]. In particular, intelligent assistants and conversational agents are expected to recognize emotional tone and generate responses that align with users' affective states. However, relying solely on textual input is often insufficient, as human emotions are subtle, overlapping, and highly context-dependent, making them difficult to accurately capture using unimodal signals [3]. Furthermore, external factors such as background noise, variations in speaker tone, and environmental conditions introduce additional challenges for emotion recognition systems [3]. As a result, multimodal approaches that combine text and speech have gained attention due to their ability to capture complementary emotional cues from both linguistic and paralinguistic information. While visual signals, such as

facial expressions, can further enhance emotion recognition performance, their practical adoption remains limited due to challenges in large-scale data collection, annotation, and real-world deployment constraints, including privacy concerns and user preferences [4]. In many real-world applications, users are more inclined to provide audio rather than video input, making audio–text multimodal systems a more feasible and scalable solution.

Emotion recognition has attracted increasing research attention in recent years, leading to significant advancements in both unimodal and multimodal learning frameworks. Early studies primarily focused on unimodal emotion recognition using either textual information [5, 6, 7] or speech signals through handcrafted acoustic features [8, 2, 9] and, more recently, self-supervised learned speech representations [10, 11]. Although these approaches achieved promising performance, relying on a single modality often limits the ability of the model to capture the complete emotional context. To overcome these limitations, recent research has increasingly shifted toward multimodal emotion recognition by integrating complementary information from speech and text modalities. Existing multimodal approaches employ various fusion strategies, including feature interaction methods [12, 13, 14, 15, 16, 17], cross-modal attention mechanisms [18], knowledge distillation frameworks [19], and ensemble-based approaches that combine independently learned predictions or representations from different modalities [3, 20]. More recently, contrastive learning has emerged as an effective paradigm for multimodal representation learning. Existing contrastive multimodal approaches can generally be categorized into cross-modal alignment methods [21], fusion-integrated contrastive learning frameworks [22, 23, 24], and pretraining-based contrastive approaches [25, 26].

Several studies have demonstrated the effectiveness of multimodal learning and contrastive objectives for emotion recognition. For example, the ablation study presented in [22] showed that multimodal fusion significantly outperformed unimodal approaches on both the IEMOCAP and MELD datasets, achieving approximately 18% improvement compared to speech-only models and around 4% improvement compared to text-only models. Similarly, [21] reported that incorporating cross-modal contrastive alignment improved weighted accuracy on the IEMOCAP dataset by 1.44%. In addition, [23] demonstrated that integrating label-aware contrastive learning within multimodal fusion increased the weighted average F1-score on the MELD dataset by approximately 4%. These findings collectively highlight the effectiveness of combining multimodal learning with contrastive representation objectives for emotion recognition tasks.

Despite these advances, several challenges remain insufficiently addressed. Existing methods still struggle to achieve robust generalization across different datasets, speakers, and linguistic distributions, while many fusion strategies rely on static interactions that fail to adaptively model the varying reliability and importance of modalities across conversational contexts. Furthermore, current contrastive learning approaches often focus on either cross-modal alignment or discriminative representation learning independently, limiting their ability to jointly optimize semantic consistency and emotional separability within multimodal representations. To address these limitations, we propose a unified multimodal framework that integrates speech and text through a dynamic utterance-level fusion mechanism combined with structured cross-modal contrastive learning.

Specifically, the proposed framework introduces a chunk-based audio modeling strategy in which each utterance is segmented into overlapping temporal chunks to better capture fine-grained variations in emotional expression. Supervised contrastive learning is applied at the chunk level to enhance intra-class compactness and inter-class separability within the audio representation space. To further improve robustness, a confidence-aware aggregation mechanism is employed to weight chunk-level representations according to their reliability, producing a more stable utterance-level embedding. In parallel, a cross-modal contrastive objective aligns audio and text representations within a shared embedding space, improving inter-modal consistency. By jointly optimizing dynamic fusion, chunk-level supervised contrastive learning, cross-modal alignment, and confidence-aware aggregation, the proposed framework achieves improved generalization and provides a robust and discriminative solution for multimodal emotion recognition across diverse conditions.

2. Related Work

Emotion recognition has evolved significantly over recent years, progressing from traditional unimodal approaches toward more advanced multimodal and contrastive learning frameworks. Early research primarily focused on unimodal emotion recognition using either textual or acoustic information independently [5, 6, 7, 8, 2, 9, 10, 11]. Although these approaches demonstrated promising performance, relying on a single modality often limits the model's ability to capture

the full emotional context. Consequently, later studies shifted toward multimodal frameworks that integrate complementary information from different modalities through various fusion strategies [12, 13, 14, 15, 16, 17, 18, 19, 3, 20]. More recently, contrastive learning has emerged as an effective paradigm for improving multimodal representation learning by enhancing cross-modal alignment and increasing intra-class discriminability [21, 24, 22, 23, 25, 26]. Figure 1 summarizes the main directions in emotion recognition research and highlights representative studies within each category.

In text-based emotion recognition, transformer architectures have achieved strong performance in modeling contextual semantics and linguistic dependencies [5]. However, semantic understanding alone is often insufficient for accurately capturing subtle emotional cues embedded within text. To address this limitation, several studies proposed ensemble-based approaches that separately learn semantic and emotional representations before combining them for final classification [6]. Building upon this idea, subsequent research introduced attention mechanisms to better emphasize emotionally relevant contextual information, leading to improved feature representation and classification performance [7].

For speech-based emotion recognition, early approaches mainly relied on handcrafted acoustic features due to their structured and interpretable nature. These features have been widely used in both traditional machine learning and deep learning frameworks. Some studies transformed handcrafted features into two-dimensional representations to exploit the representational capabilities of visual models [8, 2], while others directly processed one-dimensional acoustic feature vectors using ensemble learning and conventional classification models [9]. Despite their effectiveness, handcrafted features remain limited in their ability to capture high-level emotional representations. To overcome these limitations, recent research has increasingly adopted self-supervised learning (SSL) for speech representation learning. Instead of relying on manually designed acoustic descriptors, SSL models learn generalized and transferable representations directly from large-scale unlabeled speech data. Comparative studies have demonstrated the superiority of these learned representations over traditional handcrafted features across emotion recognition tasks [10]. Furthermore, several works investigated different adaptation strategies, including full and partial fine-tuning, to better exploit pretrained speech representations for downstream emotional understanding [11].

The transition from unimodal to multimodal emotion recognition was primarily motivated by the limitation of relying on a single modality to capture the complete emotional context. Consequently, recent research has increasingly focused on integrating complementary information from multiple modalities, particularly speech and text, through different fusion strategies. Existing multimodal approaches can generally be categorized into feature interaction, attention-based fusion, knowledge distillation, and ensemble-based methods.

A considerable portion of prior work focuses on feature interaction mechanisms, where modalities are jointly fused to enhance representation learning. Some studies emphasize one modality more than the other during the interaction process. For instance, [12] refines textual representations using acoustic context, where audio information is utilized to disambiguate lexically neutral or ambiguous text through a multi-head attention fusion mechanism. In this framework, acoustic features serve as the keys and values, while textual representations form the query. Other approaches adopt a hybrid strategy by training independent models for each modality alongside an additional multimodal fusion model, followed by combining their predictions using ensemble learning [13]. Furthermore, several studies incorporate intermediate fusion stages between modalities before the final classification layers to strengthen multimodal representation learning [14, 15].

Beyond speech and text, some studies extend multimodal emotion recognition by incorporating the visual modality [14, 16]. These approaches extract visual representations using deep visual feature extraction techniques and subsequently integrate them with speech and textual information. For example, [16] combines multiple handcrafted acoustic features before performing multimodal fusion with visual representations. Similarly, [17] demonstrates that directly learning from raw speech and textual inputs can outperform traditional handcrafted feature extraction pipelines. Their framework combines CNNs, BiLSTM networks, self-attention, and multi-head attention modules for each modality independently, while employing multi-stage fusion strategies at different representation levels. Specifically, the model first fuses convolutional representations, followed by fusion after the self-attention BiLSTM layers, before aggregating all representations using a transformer encoder for final emotion classification.

Although attention mechanisms have been introduced in several multimodal frameworks, many studies still primarily rely on feature interaction as the core fusion philosophy. In contrast, attention-based

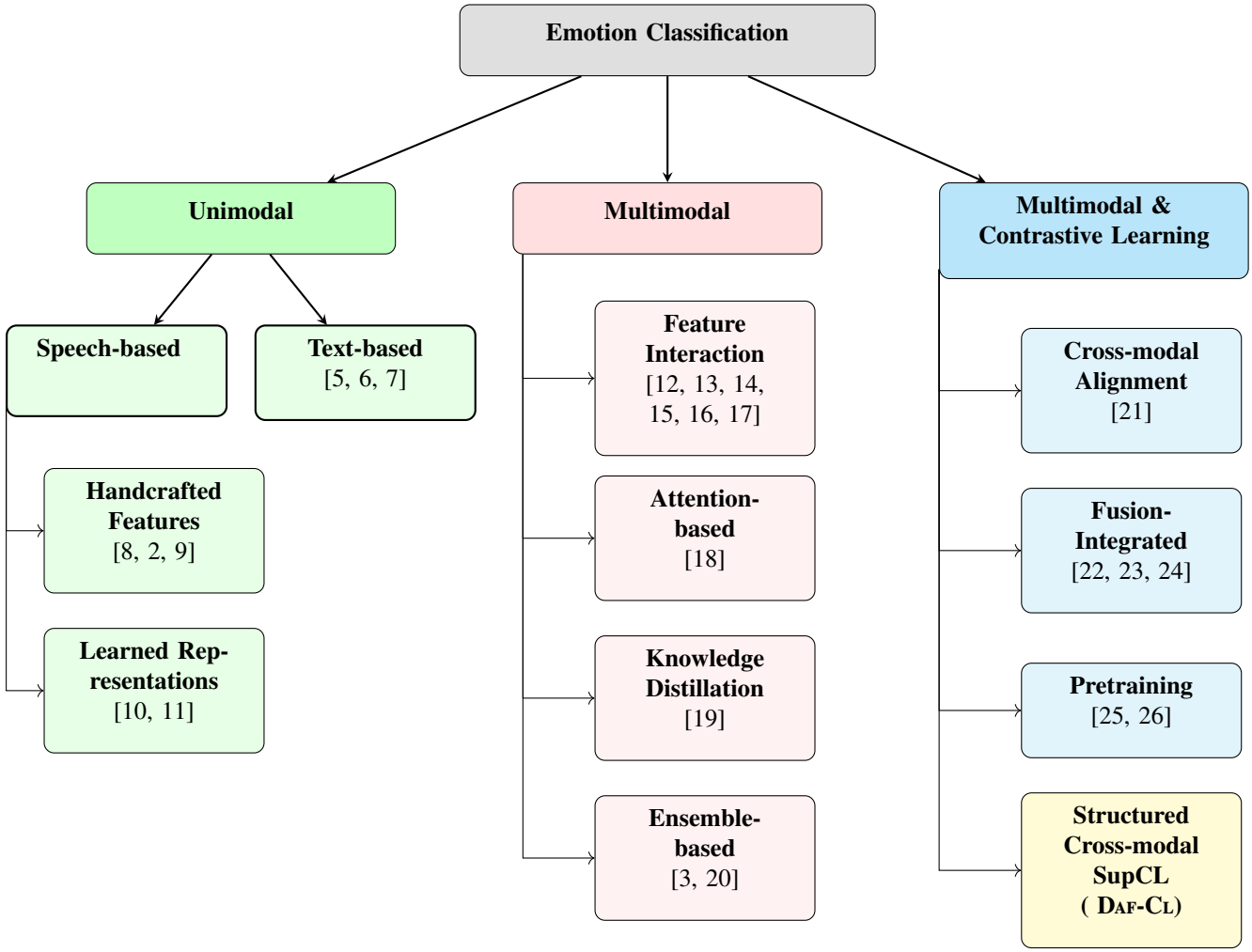


Figure 1: Taxonomy of Emotion Recognition: From Unimodal to Multimodal Contrastive Learning.

approaches explicitly model the alignment between modalities through cross-modal attention mechanisms. For example, [18] employs cross-modal attention to enhance the interaction between textual and acoustic representations for emotion recognition. The framework utilizes 1D convolutional neural networks for feature extraction, while the cross-modal attention module enables the model to focus on emotionally salient information across modalities.

Other research directions focus on improving multimodal representation learning through knowledge distillation strategies. In [19], a dual distillation framework is proposed that combines both self-distillation and cross-modal distillation for multimodal emotion recognition in conversational settings. The self-distillation module enhances intra-modal feature representations by transferring knowledge within the same modality, while the cross-modal distillation module aligns complementary emotional information between speech and text modalities, resulting in more robust and discriminative multimodal representations.

In contrast to interaction-based methods, some approaches focus on combining independently learned predictions or representations from different modalities without performing deep cross-modal interaction during feature learning. These methods are generally categorized as ensemble-based fusion approaches [3, 20]. For instance, [20] proposes an ensemble fusion framework optimized using the Aquila optimizer, where modalities are processed separately and their outputs are aggregated during the final decision stage. Similarly, [3] independently learns audio and textual representations before combining their predictions for final emotion classification, without employing explicit cross-modal interaction mechanisms.

With the rapid advancement of self-supervised and contrastive learning, recent studies have increasingly incorporated contrastive objectives into multimodal emotion recognition frameworks. Existing approaches can generally be categorized into three main directions: cross-modal alignment, fusion-integrated contrastive learning, and pretraining-based methods.

Some studies focus primarily on aligning representations from different modalities through contrastive objectives, commonly referred to as cross-modal alignment methods. For example, [21] applies contrastive regularization between speech and text representations before multimodal fusion. In this framework, audio signals are first converted into Mel-spectrograms and processed using a CNN, while textual features are extracted using GloVe embeddings followed by an LSTM encoder. The learned representations are then passed through a discriminator that maximizes the similarity between paired audio-text embeddings while separating them from negative samples, thereby improving cross-modal representation consistency prior to fusion.

Another line of research integrates discriminative objectives during or after multimodal fusion, which can be categorized as fusion-integrated contrastive learning. In [24], the framework first performs multimodal fusion using a crossmodal transformer with focus-attention, enabling speech and text representations to interact and exchange contextual information. The fused representation is subsequently refined using a metric learning objective that enhances the separation between emotional classes within the embedding space. Similarly, [22] also applies a discriminative objective after multimodal fusion; however, the underlying learning objective differs from [24]. While [24] mainly employs metric learning to optimize embedding distances and improve the geometric structure of emotional representations, [22] explicitly encourages intra-class compactness and inter-class separability by pulling samples from the same emotion class closer while pushing different emotional classes farther apart. Extending this direction, [23] proposes a multilayer interaction framework that progressively exchanges information across multimodal representations to improve emotional feature learning. Unlike methods that apply contrastive learning only after the final fusion stage, MIMCL tightly integrates contrastive objectives within intermediate interaction modules, enabling iterative refinement of multimodal representations and improving emotional discriminability across multiple feature levels.

In contrast, other studies mainly employ contrastive learning during the pretraining stage to learn more generalized multimodal representations. For instance, [25] introduces a contrastive language-audio pretraining framework for affective computing that aligns speech and textual representations using soft-target contrastive learning. Unlike conventional hard-label contrastive objectives, the proposed method leverages soft targets to capture nuanced emotional relationships between samples, resulting in more robust multimodal representations and improved downstream emotion recognition performance. Similarly, [26] proposes a contrastive language-audio alignment framework that combines pretrained HuBERT speech representations with self-alignment learning to enhance emotional representation quality. Although multimodal contrastive learning is employed during pretraining, the downstream emotion recognition stage primarily relies on the enhanced speech representations, demonstrating the effectiveness of cross-modal pretraining for speech emotion

understanding.

Our research addresses an important gap in the existing literature by introducing a Dynamic Adaptive Fusion with Contrastive Learning (DAF-CL) framework that jointly integrates dynamic cross-modal fusion and structured supervised contrastive learning within a unified optimization framework. The proposed approach aims to address several limitations that remain insufficiently explored in current emotion recognition methods. Although recent multimodal contrastive learning approaches have demonstrated promising performance, existing methods still struggle to effectively model semantic alignment, adaptive multimodal interaction, and discriminative emotional representation learning simultaneously.

One major limitation of unimodal approaches is their restricted ability to capture complementary emotional information available across different modalities [5, 6, 7, 8, 2, 9, 10, 11]. Consequently, multimodal emotion recognition has emerged as a promising direction; however, many existing multimodal frameworks still suffer from insufficient semantic alignment between heterogeneous modalities. For example, in [12], the framework primarily relies on textual representations while utilizing speech only to disambiguate lexically neutral or ambiguous text. Similarly, methods such as [3, 15] and [16] mainly employ intermediate fusion strategies without explicitly ensuring optimal interaction and alignment between modalities.

Moreover, several existing multimodal approaches rely on static fusion mechanisms, which may inadequately model the varying contribution and importance of modalities across different emotional contexts [13, 15], and [20]. In realistic conversational settings, the emotional significance of speech and text may dynamically vary across utterances, making static fusion strategies less effective for robust emotion understanding.

Although contrastive learning has recently gained significant attention in multimodal emotion recognition, existing approaches often focus on either cross-modal alignment or class-level discrimination independently rather than jointly optimizing both objectives [21, 24, 22]. Furthermore, several studies apply contrastive objectives only after the final fusion stage, limiting progressive interaction refinement and multimodal representation learning throughout the network [24, 22, 23]. On the other hand, pretraining-based approaches improve representation quality during the alignment stage but often fail to fully exploit multimodal interaction during downstream emotion classification tasks [25] and [26].

Another important limitation lies in the generalization capability of existing methods. Many studies primarily evaluate their frameworks on a limited number of benchmark datasets, particularly IEMOCAP and RAVDESS, without extensively investigating model robustness across different speakers, domains, and data distributions [26, 2, 13, 15] and [18, 19, 20, 21, 24, 22, 23]. As a result, the real-world generalizability of these models remains insufficiently explored.

To address these limitations, we propose the DAF-CL framework, which combines dynamic adaptive multimodal fusion with structured supervised contrastive learning to jointly enhance cross-modal semantic alignment and emotional discriminability. The proposed framework is designed to enable more effective multimodal interaction, adaptive modality weighting, and improved generalization across diverse emotional contexts and data distributions, providing a scalable and robust solution for multimodal emotion recognition.

3. DAF-CL Model Development

In this section, we present the proposed **Dynamic Adaptive Fusion with Contrastive Learning (DAF-CL)** framework for multimodal emotion recognition. Given a multimodal utterance set denoted by ($\mathcal{D}$), the framework jointly models textual and acoustic information, represented as:

$$\mathcal{D} = \{(t_i, u_i, y_i)\}_{i=1}^{N} \tag{1}$$

where t_i and u_i denote the textual and speech utterances of the ith sample, respectively, y_i represents the corresponding emotion label, and N is the total number of utterances in the dataset.

The proposed DAF-CL framework consists of three major components. First, a structured supervised contrastive learning module is applied to overlapping speech chunks to enhance intra-class compactness and inter-class separability within the acoustic representation space. Second, a dynamic confidence-aware audio-text fusion mechanism is introduced to adaptively integrate multimodal information according to the contextual reliability of each modality. Third, a cross-modal contrastive alignment module is employed to improve semantic consistency between speech and textual representations by aligning heterogeneous modality embeddings within a shared representation space.

To jointly optimize these components, the framework is trained using a multi-objective learning strategy that combines supervised contrastive loss, cross-modal alignment loss, and emotion classification loss. The

overall architecture is illustrated in Figure 2. The following subsections describe each component in detail.

3.1. Audio Representation Learning

Given an input speech utterance waveform u_i, we segment the signal into overlapping, fixed-length chunks to capture local emotional variations while maintaining a consistent input size for the downstream models. The main notations used throughout this subsection are summarized in Table 1.

Table 1: Notation summary for audio representation learning.

Notation	Description
M	Number of chunks per utterance.
c	Chunk duration in seconds.
f_s	Audio sampling rate.
ρ	Overlap ratio ($\rho \in [0, 1)$).
$\mathbf{X}_a^{(i,m)}$	Extracted acoustic features for chunk m of utterance i.
L	Total number of WavLM Transformer layers .
$\mathbf{H}_i^{(l,m)}$	Hidden representation of chunk m at layer l, $\mathbf{H}_i^{(l,m)} \in \mathbb{R}^{T_m \times d}$.
T_m	Temporal frame length of chunk m in the latent space.
d	Feature embedding dimension.
w_l	Trainable importance weight assigned to layer l.
ω_l	Softmax-normalized layer importance weight.

The corresponding chunk length (L_c) and hop size (L_h) in terms of samples are defined as:

$$L_c = \lfloor c \cdot f_s \rfloor, \quad L_h = \lfloor L_c(1 - \rho) \rfloor. \tag{2}$$

The segmentation is performed using a sliding-window approach. The m-th chunk ($m \in \{1, \ldots, M\}$) of the utterance waveform is extracted as:

$$u_i^{(m)} = u_i[(m-1)L_h : (m-1)L_h + L_c]. \tag{3}$$

For utterances shorter than L_c, a single chunk is produced and zero-padded to match the fixed length L_c. For longer utterances, the window strides by L_h, and the final segment is padded if necessary. This strategy ensures fixed-length, overlap-aware acoustic segmentation.

Each chunk $u_i^{(m)}$ is processed by a pretrained WavLM to extract hierarchical acoustic features across multiple Transformer layers. To preserve the general acoustic knowledge learned during self-supervised pretraining while adapting to downstream emotion dynamics, the lower layers of WavLM are frozen, whereas the higher layers are fine-tuned.

Unlike conventional approaches that rely solely on the final layer representation, the proposed framework introduces a trainable layer fusion mechanism to adaptively aggregate information across all L layers. We define a vector of trainable layer weights:

$$\mathbf{w} = [w_1, w_2, \ldots, w_L], \tag{4}$$

which are normalized via a softmax function to produce the layer-wise attention scores:

$$\omega_l = \frac{\exp(w_l)}{\sum_{k=1}^{L} \exp(w_k)}. \tag{5}$$

The layer-fused representation for chunk m of utterance i is then computed as:

$$\mathbf{H}_{\text{fused}}^{(i,m)} = \sum_{l=1}^{L} \omega_l \mathbf{H}_i^{(l,m)}. \tag{6}$$

This adaptive fusion mechanism enables the model to selectively emphasize specific Transformer layers that capture emotionally discriminative acoustic cues at different abstraction levels.

To model the localized temporal dependencies within each chunk, the fused representation is passed through a vanilla Transformer encoder layer:

$$\widetilde{\mathbf{H}}^{(i,m)} = \text{TransformerEncoder}(\mathbf{H}_{\text{fused}}^{(i,m)}). \tag{7}$$

The frame-level representations are subsequently aggregated into a fixed-dimensional chunk embedding $\mathbf{z}_i^{(m)} \in \mathbb{R}^d$ using mean pooling across the temporal dimension:

$$\mathbf{z}_i^{(m)} = \frac{1}{T_m} \sum_{t=1}^{T_m} \widetilde{\mathbf{H}}_t^{(i,m)}, \tag{8}$$

where $\widetilde{\mathbf{H}}_t^{(i,m)}$ denotes the t-th temporal frame of the encoded chunk.

Since different speech segments contribute unequally to the underlying emotional state, an adaptive reliability assessment mechanism is introduced to estimate the contribution of each acoustic chunk to the utterance-level prediction. The overall confidence-aware chunk weighting and aggregation process is illustrated in figure 3. Specifically, each chunk is assigned a learned emotional saliency score and a prediction-confidence score, which are combined to determine its relative contribution during weighted aggregation. First, an emotional saliency score $\alpha_i^{(m)}$ is computed to estimate the emotional relevance of the chunk:

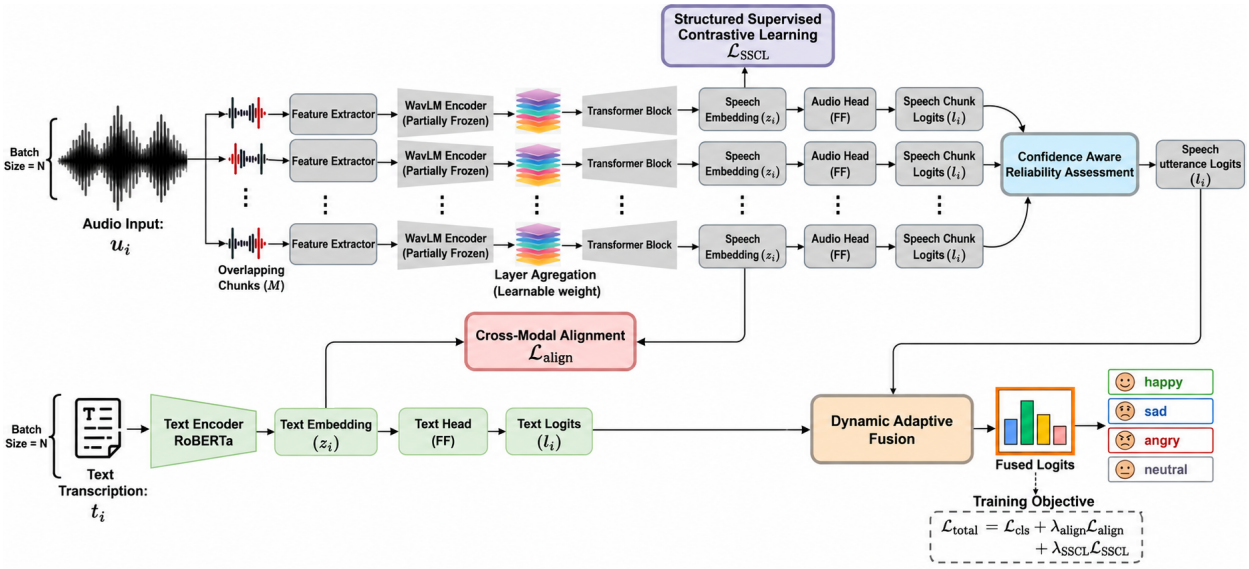


Figure 2: Framework development of the DAF-CL model.

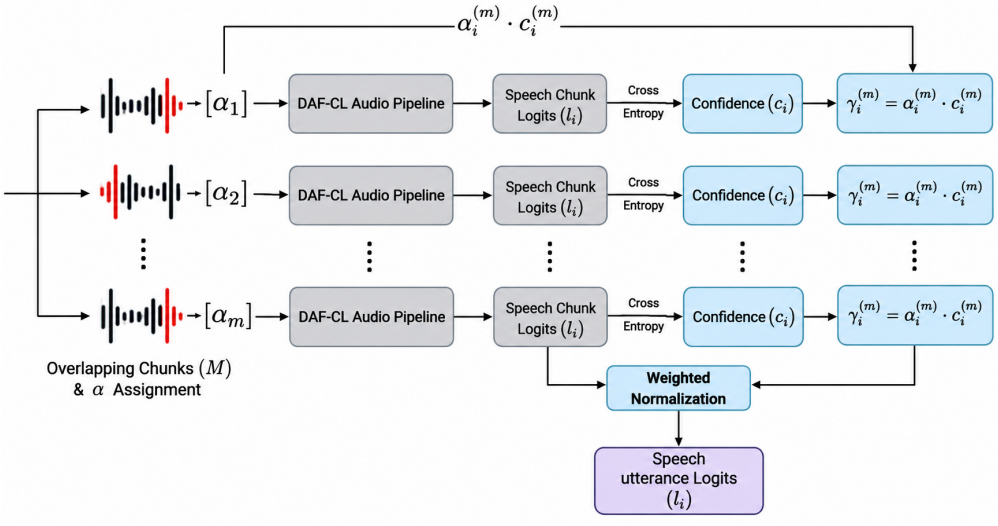


Figure 3: Confidence-aware reliability assessment and aggregation of acoustic chunks.

$$\alpha_i^{(m)} = \sigma(\mathbf{W}_a \mathbf{z}_i^{(m)} + \mathbf{b}_a), \tag{9}$$

where $\sigma(\cdot)$ is the sigmoid activation function, and $\mathbf{W}_a \in \mathbb{R}^{1\times d}$ and $\mathbf{b}_a \in \mathbb{R}^1$ are learnable parameters.

Each chunk embedding is passed through the audio classification head $f_{\text{audio}}(\cdot)$ to compute the chunk-level emotion prediction logits $\ell_i^{(m)}$:

$$\ell_i^{(m)} = f_{\text{audio}}(\mathbf{z}_i^{(m)}). \tag{10}$$

The model's local prediction confidence is estimated from the uncertainty of the chunk-level emotion distribution. First, the chunk logits are converted into class probabilities:

$$\mathbf{p}_i^{(m)} = \text{softmax}\left(\ell_i^{(m)}\right). \tag{11}$$

The predictive uncertainty of the m-th chunk is quantified using Shannon entropy:

$$\mathcal{E}_i^{(m)} = -\sum_{k=1}^{C} p_{i,k}^{(m)} \log\left(p_{i,k}^{(m)} + \epsilon\right), \tag{12}$$

where C denotes the number of emotion classes and $\epsilon > 0$ is a small constant introduced for numerical stability. The entropy value is subsequently converted into an inverse-entropy confidence score, such that chunks with lower predictive uncertainty receive higher confidence. To stabilize its relative scale across the processed chunk pool, the resulting confidence values are normalized by their mean and clipped to the interval $[0.1, 10.0]$.

The final chunk importance weight $\gamma_i^{(m)}$ combines the learned emotional saliency with the normalized predictive confidence:

$$\gamma_i^{(m)} = \alpha_i^{(m)} \cdot c_i^{(m)}. \tag{13}$$

Finally, the utterance-level decision vector ℓ_i is achieved via a weight-normalized decision fusion step across all acoustic segments:

$$\ell_i^{\text{audio}} = \frac{\sum_{m=1}^{M} \gamma_i^{(m)} \ell_i^{(m)}}{\sum_{m=1}^{M} \gamma_i^{(m)} + \epsilon}, \tag{14}$$

where $\epsilon > 0$ is a small constant preventing numerical instability. This decision-level fusion mechanism adaptively emphasizes predictions from emotionally clear speech regions while suppressing the influence of ambiguous or noisy chunks on the final utterance classification.

3.2. Text Representation Learning

Given the text transcription associated with the i-th utterance, denoted by t_i, the framework leverages a pretrained RoBERTa encoder to capture semantic and contextual dependencies across the word tokens.

Following standard practice in RoBERTa-based language models, the contextual hidden vector corresponding to the special start-of-sentence token (`<s>`) is adopted as the utterance-level textual embedding:

$$\mathbf{z}_i^{\text{text}} = \mathbf{h}_{i,\texttt{<s>}} \tag{15}$$

where $\mathbf{z}_i^{\text{text}} \in \mathbb{R}^{d_{\text{text}}}$ serves as the fixed-dimensional semantic summary of the transcription sequence.

To obtain text-specific emotion predictions, this embedding is forwarded through a task-specific classification head $f_{\text{text}}(\cdot)$ consisting of fully connected layers, ReLU activation, and dropout regularization to produce the text emotion logits:

$$\ell_i^{\text{text}} = f_{\text{text}}\left(\mathbf{z}_i^{\text{text}}\right) \tag{16}$$

where $\ell_i^{\text{text}} \in \mathbb{R}^C$ represents the text-specific emotion logits over the C target classes. Both $\mathbf{z}_i^{\text{text}}$ and ℓ_i^{text} serve as the core textual constituents for the subsequent cross-modal alignment and multimodal fusion stages.

3.3. Cross-Modal Alignment

Although the acoustic and textual representations are derived from the same utterance, they originate from heterogeneous feature spaces and may exhibit modality-specific structural discrepancies. To improve their compatibility prior to multimodal fusion, the proposed framework aligns the two modalities through a shared latent representation, as illustrated in figure 4. Specifically, the acoustic and textual embeddings are first transformed by modality-specific projection heads and mapped into a common d-dimensional space:

$$\mathbf{v}_i^{\text{audio}} = f_a(\mathbf{z}_i), \qquad \mathbf{v}_i^{\text{text}} = f_t(\mathbf{z}_i^{\text{text}}) \tag{17}$$

where $f_a(\cdot)$ and $f_t(\cdot)$ denote learnable projection layers, while $\mathbf{v}_i^{\text{audio}}, \mathbf{v}_i^{\text{text}} \in \mathbb{R}^d$ represent the aligned audio and text embeddings of the i-th utterance, respectively. To ensure stable similarity calculations, these vectors are L_2-normalized such that $\|\mathbf{v}_i^{\text{audio}}\|_2 = \|\mathbf{v}_i^{\text{text}}\|_2 = 1$.

To encourage semantic consistency across the modalities, a bidirectional contrastive learning objective is employed. For a mini-batch of size N, a pairwise similarity matrix $\mathbf{S} \in \mathbb{R}^{N\times N}$ is computed, where the entry S_{ij} measures the scaled semantic proximity between the

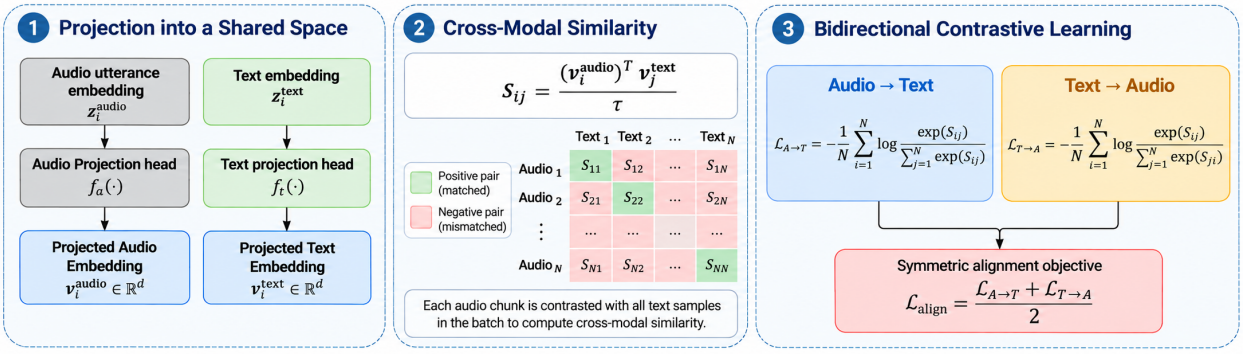


Figure 4: Overview of the proposed cross-modal audio–text alignment mechanism

audio representation of utterance i and the textual representation of utterance j:

$$S_{ij} = \tau \cdot (\mathbf{v}_i^{\text{audio}})^{\top} \mathbf{v}_j^{\text{text}} \tag{18}$$

where τ is a learnable temperature scaling parameter.

The audio-to-text contrastive loss ($\mathcal{L}_{A\to T}$) for the mini-batch is formulated using the InfoNCE objective as:

$$\mathcal{L}_{A\to T} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp(S_{ii})}{\sum_{j=1}^{N} \exp(S_{ij})} \tag{19}$$

Conversely, the text-to-audio contrastive loss ($\mathcal{L}_{T\to A}$) captures the symmetry of the alignment space and is defined as:

$$\mathcal{L}_{T\to A} = -\frac{1}{N}\sum_{i=1}^{N} \log \frac{\exp(S_{ii})}{\sum_{j=1}^{N} \exp(S_{ji})} \tag{20}$$

The overall cross-modal alignment objective is achieved by optimizing the joint bidirectional contrastive loss:

$$\mathcal{L}_{\text{align}} = \frac{\mathcal{L}_{A\to T} + \mathcal{L}_{T\to A}}{2} \tag{21}$$

By maximizing the similarity of true matched audio-text pairs (S_{ii}) while simultaneously minimizing the cross-modal similarity of mismatched negative pairs (S_{ij} where $i \neq j$), this alignment strategy forces the model to learn modality-invariant representations. Consequently, semantically corresponding cross-modal features are pulled into compact joint clusters, significantly reducing the semantic gap and preparing the representations for downstream multimodal fusion.

3.4. Dynamic Adaptive Fusion

Conventional fusion strategies typically assign fixed contributions to individual modalities, which can be suboptimal since the reliability of acoustic and textual cues inherently varies across different utterances. To address this limitation, a dynamic adaptive fusion mechanism is introduced to automatically adjust the contribution of each modality based on the contextual characteristics of the input utterance as illustrated in figure 5 .

The proposed framework derives an utterance-level acoustic gating coefficient from the learned chunk-level saliency scores. Specifically, for the i-th utterance, the modality fusion coefficient $\lambda_i \in [0, 1]$ is obtained by averaging the learned saliency scores across its M acoustic segments:

$$\lambda_i = \frac{1}{M_i}\sum_{m=1}^{M} \alpha_i^{(m)} \tag{22}$$

Here, M denotes the learned emotional saliency of the m-th chunk. In contrast, the confidence-gated weight $\gamma_i^{(m)} = \alpha_i^{(m)} c_i^{(m)}$ is used specifically for aggregating chunk-level audio predictions into the utterance-level audio logits. This separation allows predictive confidence to determine the relative importance of individual acoustic segments, while the learned saliency scores determine the overall contribution of the acoustic modality during multimodal fusion.

The final multimodal classification decision is subsequently achieved through an adaptive late fusion of the modality-specific prediction logits:

$$\ell_i^{\text{final}} = \lambda_i \ell_i^{\text{audio}} + (1 - \lambda_i)\ell_i^{\text{text}} \tag{23}$$

where ℓ_i^{audio} represents the weight-aggregated audio logits, and ℓ_i^{text} represents the textual logits. Under this formulation, higher values of λ_i dynamically scale up the contribution of the acoustic stream when the speech signal is deemed reliable, whereas lower values automatically shift the model's emphasis toward the textual modality.

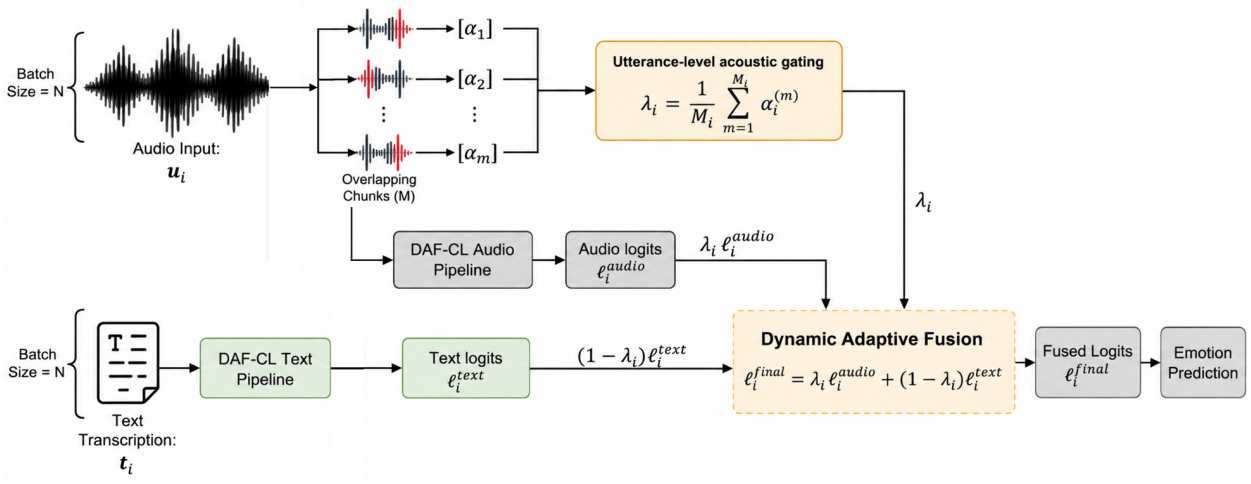


Figure 5: Overview of the proposed dynamic adaptive fusion mechanism.

Unlike conventional gating mechanisms that introduce a separate modality-level gating network, the proposed strategy derives the fusion coefficient directly from the learned chunk-level acoustic saliency scores. The local confidence term is used independently during acoustic decision aggregation, where it modulates the contribution of each chunk according to the certainty of its emotion prediction. Consequently, the framework establishes a hierarchical weighting strategy: confidence-aware chunk weighting determines the utterance-level acoustic decision, while the aggregated saliency scores govern the subsequent balance between the acoustic and textual modalities.

3.5. Structured Supervised Contrastive Learning

To enforce an emotionally structured topology in the latent space, we introduce a Structured Supervised Contrastive Learning (SSCL) mechanism, whose overall processing flow is illustrated in figure 6. Unlike standard Supervised Contrastive Learning (SupCon), which relies on a binary distinction between positive (same-class) and negative (different-class) pairs, the proposed SSCL incorporates a continuous, class-aware similarity metric. As shown in the figure, the framework first projects the acoustic chunk embeddings into a normalized latent space, computes pairwise chunk similarities, and then assigns continuous pairwise weights according to the predefined emotion semantic similarity matrix $\mathbf{S}$. These class-aware weights are subsequently incorporated into the contrastive objective, allowing the model to preserve graded relationships among acoustic chunk embeddings according to the semantic proximity of their corresponding emotion classes while maintaining discriminative structure in the latent space.

Let $\mathbf{S} \in \mathbb{R}^{C\times C}$ denote a predefined emotion similarity matrix, where C represents the total number of target emotion classes. Each element $s_{ab} \in [0, 1]$ encodes the relative affective similarity between emotion classes a and b. Rather than interpreting these values as absolute psychological measurements, $\mathbf{S}$ is designed as a fixed structured prior that specifies the relative proximity among the target emotion categories. For the four emotion classes considered in this work, ordered as Sad, Neutral, Happy, and Angry, the similarity matrix is defined as:

$$\mathbf{S} = \begin{bmatrix} 1.0 & 0.6 & 0.2 & 0.5 \\ 0.6 & 1.0 & 0.7 & 0.4 \\ 0.2 & 0.7 & 1.0 & 0.3 \\ 0.5 & 0.4 & 0.3 & 1.0 \end{bmatrix}. \tag{24}$$

The diagonal elements are set to 1.0, corresponding to maximum similarity between samples belonging to the same emotion class, whereas the off-diagonal elements encode graded inter-class relationships. Comparatively higher values are assigned to emotion pairs with closer affective characteristics, such as Neutral–Happy and Sad–Neutral, while lower values are assigned to more distinct emotion pairs, such as Sad–Happy. Sad–Angry is assigned an intermediate similarity since both emotions share negative affective characteristics while differing in activation. Accordingly, the numerical values are used to encode the relative ordering of inter-emotion similarity rather than exact psychological distances. The effectiveness of this structured prior is further evaluated experimentally through comparison with

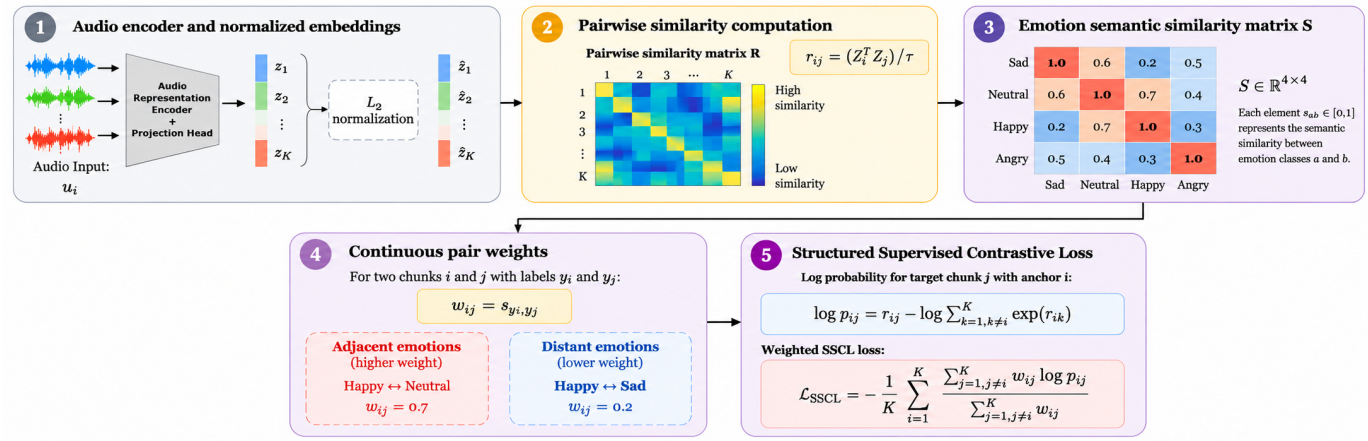


Figure 6: Overview of the proposed Structured Supervised Contrastive Learning (SSCL) mechanism.

conventional supervised contrastive learning.

For a mini-batch of N utterances, where utterance i is partitioned into M_i acoustic segments, the total pool of processed chunks evaluated by the contrastive loss is defined as:

$$K = \sum_{i=1}^{N} M_i. \tag{25}$$

Let $\mathbf{z}_i, \mathbf{z}_j \in \mathbb{R}^d$ represent the L_2-normalized embeddings of two distinct chunks within this combined pool, and let $y_i, y_j \in \{1, \ldots, C\}$ denote their respective ground-truth emotion labels. The pairwise log-bilinear similarity score r_{ij} between anchor chunk i and contrastive target chunk j is scaled by a temperature parameter τ:

$$r_{ij} = \frac{\mathbf{z}_i^\top \mathbf{z}_j}{\tau} \tag{26}$$

For any pair of chunk samples i and j, a continuous semantic scaling weight w_{ij} is directly indexed from the emotion similarity matrix according to their class assignments:

$$w_{ij} = s_{y_i,y_j} \tag{27}$$

Consequently, pairs associated with more closely related emotion categories receive larger semantic weights, whereas pairs belonging to more distant emotion categories receive smaller weights.

Because the semantic similarity matrix assigns non-zero scalar weights across all class combinations, the traditional binary separation into distinct positive and negative sets is generalized into a continuous spectrum of inter-class relationships. Each item in the contrastive batch pool therefore contributes to the anchor according to its class-dependent semantic similarity. The structured log-probability of target chunk j relative to anchor chunk i is formulated as:

$$\log p_{ij} = r_{ij} - \log \sum_{k=1,k\neq i}^{K} \exp(r_{ik}) \tag{28}$$

The proposed Structured Supervised Contrastive Loss ($\mathcal{L}_{\text{SSCL}}$) across the entire contrastive batch pool is formulated as the weighted expectation:

$$\mathcal{L}_{\text{SSCL}} = -\frac{1}{K}\sum_{i=1}^{K} \frac{\sum_{j=1,j\neq i}^{K} w_{ij} \log p_{ij}}{\sum_{j=1,j\neq i}^{K} w_{ij}} \tag{29}$$

By replacing discrete class masking with matrix-guided continuous weighting, the proposed objective encourages the acoustic embedding space to preserve graded inter-emotion relationships while maintaining discriminability among the target emotion categories. In this way, SSCL complements the cross-modal alignment objective by introducing class-aware structure within the acoustic representation space rather than directly altering the final emotion labels or multimodal decision rule.

3.6. Joint Optimization Objective

The final stage of the proposed framework involves multi-task learning to simultaneously optimize classification accuracy, cross-modal alignment, and structural latent topology. After performing the dynamic adaptive fusion on the modality-specific prediction vectors, the final utterance-level emotion prediction is determined by applying an argmax operation over the fused logit vector ℓ_i^{final}.

To train the network, the classification loss $\mathcal{L}_{\text{cls}}$ is formulated using the standard Cross-Entropy loss over a

mini-batch of size N:

$$\mathcal{L}_{\text{cls}} = -\frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C} y_{i,c}\log\left(\text{softmax}(\ell_i^{\text{final}})_c\right) \tag{30}$$

where $y_{i,c}$ is a binary indicator indicating whether class label c is the ground-truth classification for the i-th utterance, and C represents the total number of emotion categories.

To leverage the complementary advantages of feature-level alignment and structured semantic clustering, the overall network is optimized end-to-end using a joint objective function. The total loss $\mathcal{L}_{\text{total}}$ is defined as a weighted linear combination of the classification loss, the bidirectional cross-modal alignment loss, and the structured supervised contrastive loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{cls}} + \lambda_{\text{align}}\mathcal{L}_{\text{align}} + \lambda_{\text{SSCL}}\mathcal{L}_{\text{SSCL}} \tag{31}$$

where $\lambda_{\text{align}} > 0$ and $\lambda_{\text{SSCL}} > 0$ are regularizing hyperparameters that control the relative trade-offs and contributions of the cross-modal alignment and structured embedding objectives, respectively. By minimizing $\mathcal{L}_{\text{total}}$, the framework guarantees that the learned representations are concurrently discriminative for downstream classification, aligned across text and audio spaces, and topologically organized according to domain-specific emotional semantics.

4. Performance Evaluation

To evaluate the proposed DAF-CL framework, we conduct a systematic empirical evaluation centered on the macro F_1-score. Our experimental design investigates classification accuracy, architectural contribution, hyperparameter stability, and latent representation topology across five analytical stages detailed below (Sections 4.1–4.6).

4.1. Datasets and Preprocessing

To evaluate the proposed DAF-CL framework across diverse speakers, acoustic environments, and speaking styles, we constructed a unified multimodal benchmark corpus targeting a standard four-class speech emotion recognition task (*happy*, *sad*, *angry*, and *neutral*). The composite dataset integrates seven publicly available speech corpora: MELD [27], CREMA-D [28], IEMOCAP [29], Morgan Emotional Speech Set (MESS) [30], TESS [31], ASVP-ESD [32], and JL Corpus [33]. As detailed in Table 2, the resulting corpus aggregates 22,790 utterances ($\sim$ 20 hours of speech), balancing prosodically controlled read speech with spontaneous conversational interactions.

Table 2: Overview of the individual emotional speech datasets comprising the unified multimodal corpus.

Dataset	**Native Modality**	**Samples (Hours)**	**Speech Type**
MELD	Audio + Text	8,200 (7.2 h)	Conversational
CREMA-D	Audio + Text	4,900 (3.5 h)	Read Speech
IEMOCAP	Audio + Text	4,490 (5.6 h)	Conversational
MESS	Audio + Text	1,800 (1.45 h)	Read Speech
TESS	Audio	1,600 (1.0 h)	Read Speech
ASVP-ESD	Audio	1,000 (0.8 h)	Conversational
JL Corpus	Audio	800 (0.5 h)	Read Speech

Because our late fusion architecture relies on bimodal audio-text representations, establishing complete textual alignment across all heterogeneous sources was essential. While MELD, CREMA-D, IEMOCAP, and MESS supply native ground-truth transcriptions, the remaining datasets (TESS, ASVP-ESD, and JL Corpus) contain unannotated audio streams. To bridge this modality gap, we utilized the Whisper-large-v3 automatic speech recognition (ASR) system [34] to generate high-fidelity transcripts for all unimodal speech samples prior to textual encoding via RoBERTa.

Beyond text generation, specialized acoustic preprocessing was required to remove non-verbal vocalizations that impede speech emotion feature extraction. Specifically, spontaneous conversational recordings in MELD frequently contain high-energy background laughter segments, which corrupt spectral pitch contours and energy distributions. To address this, we deployed the Whisper Speech Event Detection (Whisper-SED) model [35] to detect and excise laughter frames, isolating pristine speech signals before chunking and acoustic feature extraction.

To ensure rigorous performance evaluation without data leakage, the processed corpus was partitioned using a speaker-independent strategy combined with emotion-stratified sampling to preserve class distributions. A strict 10% subset of the corpus was held out as an unseen test set. The remaining 90% of the data was designated for model development, where 10% was reserved as a fixed validation set for monitoring training and early stopping, and 80% was used across a 5-fold cross-validation scheme. An independent DAF-CL model was trained on each fold. During evaluation on the unseen test set, final prediction logits were derived by averaging the soft outputs across all five fold-models, minimizing prediction variance and maximizing decision ro-

bustness.

4.2. Experimental Setup and Implementation Details

The proposed DAF-CL framework was implemented using PyTorch and PyTorch Lightning, with end-to-end training conducted on two NVIDIA A100 GPUs. Following the experimental partition protocol outlined in Section 4.1, each fold-model was trained for 10 epochs. For model selection, the checkpoint achieving the highest validation Macro F_1-score on each fold was retained to construct the final five-model ensemble for test-time inference.

All input audio signals were uniformly resampled to 16,000 Hz and truncated to a maximum length of 10 seconds to bound memory consumption. For localized feature extraction, audio signals were segmented into $T_{\text{chunk}} = 3.0$ second chunks using a sliding window with a 50% overlap (1.5 seconds hop length); short chunks at sequence boundaries were zero-padded to maintain shape uniformity. Textual transcripts were processed using a pre-trained RoBERTa tokenizer, capping the sequence length at 64 tokens per utterance.

The multi-layer acoustic backbone was initialized from pre-trained `WavLM_base_plus` [36] weights, while the textual branch was initialized from an emotion-tuned RoBERTa checkpoint [37]. To preserve fundamental low-level acoustic representations while adapting higher-level contextual layers, the first four Transformer encoder layers of WavLM were kept frozen (layer_index < 4), leaving subsequent layers fully trainable. The layer-aggregation weights ω_l were uniformly initialized across all encoder layers. The temporal aggregation module was configured as a 2-layer Transformer Encoder with hidden dimension $D = 768$, 8 attention heads, feed-forward dimension 2048, and a dropout rate of 0.2. Acoustic and textual projection heads mapped their respective embeddings into a shared $D_{\text{proj}} = 256$ dimensional contrastive latent space.

End-to-end model optimization was conducted using the AdamW optimizer [38] with a weight decay parameter of 0.005 and a batch size of 4. To account for disparate convergence dynamics across sub-networks, we adopted a discriminative learning rate strategy: $\eta_{\text{enc}} = 2 \times 10^{-5}$ for the acoustic backbone, $\eta_{\text{text}} = 5 \times 10^{-5}$ for the textual encoder, and $\eta_{\text{dec}} = 1 \times 10^{-4}$ for the fusion modules and projection heads. Class imbalance was addressed by incorporating class-frequency inverse weights into the cross-entropy classification loss ($\mathcal{L}_{\text{ce}}$). Finally, the multi-task objective weights were fixed to $\lambda_{\text{align}} = 0.4$ and $\lambda_{\text{sscl}} = 0.3$, with the temperature parameter set to $\tau = 0.07$ for the structured supervised contrastive learning loss.

4.3. Results and Discussion

In this section, we present a comprehensive empirical evaluation of the proposed DAF-CL framework. We first evaluate the model's primary classification performance on the unified test set, analyzing class-wise accuracy and confusion patterns. Next, we benchmark DAF-CL against state-of-the-art multimodal contrastive learning methods on the standard IEMOCAP and MELD datasets. Finally, we assess the out-of-distribution robustness and cross-lingual generalization of our approach across three completely unseen external datasets.

4.3.1. Performance on the Test Set

Following training, the DAF-CL model was evaluated on the held-out test set described in Section 4.1. The model achieved an overall macro-averaged F_1-score of 87.37%. As illustrated in the confusion matrix in Figure 7, the framework correctly classifies the vast majority of test instances across all target categories. However, slight confusion is observable between the *happy* and *neutral* classes.

As detailed in the classification report in Table 3, the *angry* ($F_1 = 89.28\%$) and *sad* ($F_1 = 88.93\%$) categories achieved the highest classification performance, whereas *happy* ($F_1 = 86.04\%$) and *neutral* ($F_1 = 85.24\%$) yielded comparatively lower scores. These results confirm that while the model maintains high overall stability, distinguishing subtle, prosodically overlapping expressions (such as mild happiness versus neutral tone) remains an inherent acoustic challenge.

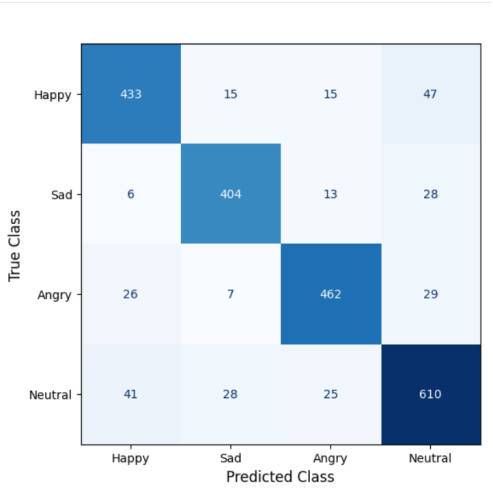


Figure 7: Confusion matrix of the DAF-CL model on the held-out test set.

Table 3: DAF-CL performance breakdown across emotion categories.

Class	Precision (%)	Recall (%)	F_1-Score (%)
Happy	85.43	86.65	86.04
Sad	89.71	88.71	88.93
Angry	88.99	89.58	89.28
Neutral	85.57	84.90	85.24
Macro Avg.	**87.42**	**87.46**	**87.37**

Table 4: Comparative performance of DAF-CL against recent contrastive learning baselines on the IEMOCAP benchmark.

Method	Weighted Acc. (%)	Unweighted Acc. (%)	Weighted F_1 (%)
Qian et al. [21]	76.06	77.45	–
Shi et al. [22]	68.77	–	68.66
Kim et al. [24]	77.40	77.70	–
Nguyen et al. [26]	77.22	–	–
DAF-CL (Proposed)	**81.50**	**82.92**	**81.88**

4.3.2. Comparison with Related Work

To contextualize the performance of DAF-CL, we compared our framework against recent state-of-the-art multimodal emotion recognition methods utilizing contrastive learning strategies [21, 24, 22, 26]. For a rigorous comparison, evaluation was conducted on two standard benchmarks: IEMOCAP [29] and MELD [27]. Tables 4 and 5 summarize the comparative findings on IEMOCAP and MELD, respectively, reporting exact metrics as published in prior works to ensure transparency.

As shown in Tables 4 and 5, DAF-CL demonstrates substantial gains over existing contrastive learning architectures. On IEMOCAP, DAF-CL surpasses the highest previously reported weighted accuracy [24] by 4.10 percentage points and unweighted accuracy by 5.22 percentage points. On the conversational MELD dataset,

Table 5: Comparative performance of DAF-CL against contrastive learning baselines on the MELD dataset.

Method	Accuracy (%)	Weighted F_1 (%)
Shi et al. [22]	65.62	64.73
DAF-CL (Proposed)	**81.66**	**81.55**

Table 6: Zero-shot generalization performance of DAF-CL on unseen external datasets.

Dataset	Emotions Evaluated	Samples	F_1 (%)
Emotional-EMA	Happy, Sad, Angry, Neutral	680	87.00
EmoV-DB-sorted	Angry, Neutral	2,836	93.00
Emo-DB (German)	Happy, Sad, Angry, Neutral	339	90.00

DAF-CL achieves a dramatic increase of over 16 percentage points in weighted F_1-score compared to recent cross-modal contrastive approaches [22]. These consistent improvements highlight the effectiveness of integrating dynamic adaptive chunk-gating with structured supervised contrastive regularization.

4.3.3. Generalization to Unseen Datasets

To evaluate out-of-distribution robustness, we evaluated the trained DAF-CL model directly on three completely unseen external datasets without fine-tuning: Emotional-EMA [39], EmoV-DB-sorted [40], and Emo-DB [41]. As summarized in Table 6, the framework maintains strong performance across varying recording environments, acoustic setups, and speaker identities.

Crucially, DAF-CL achieved a 90.00% F_1-score on the German Emo-DB corpus, despite being trained primarily on English speech. This indicates that our acoustic chunking and structured latent space capture language-invariant emotional acoustics, establishing a foundation for cross-lingual emotion recognition.

4.4. Ablation Study

To systematically evaluate the individual contributions and design choices of the proposed DAF-CL framework, we conduct a two-part empirical evaluation under identical training conditions. First, we perform a progressive ablation study to observe how each key component incrementally contributes to the overall macro F_1-score on the unified test set (Section 4.1). Second, we investigate alternative architectural design choices across four core modules: chunk aggregation, multimodal fusion, layer aggregation, and the structured contrastive loss matrix.

4.4.1. Progressive Ablation Study

We construct a simplified multimodal baseline and progressively integrate each proposed module until the full architecture is realized. The baseline utilizes only the final hidden state of WavLM (omitting adaptive layer fusion), aggregates chunk-level features via uniform mean pooling, and combines audio and text modalities using late fusion with fixed equal weighting (0.5/0.5). Both the cross-modal alignment loss and the structured supervised contrastive learning (SSCL) objective are omitted from this baseline.

Starting from this baseline, we incrementally incorporate:

1. *Adaptive Layer Fusion*: Replaces the static final WavLM hidden layer with a learned, weighted combination of all intermediate hidden layers;
2. *Chunk Weighting*: Replaces uniform mean pooling with a dynamic chunk aggregation mechanism driven by acoustic saliency gate α and classifier confidence;
3. *Dynamic Fusion*: Replaces fixed late fusion with an input-dependent gating coefficient efficiently reused from chunk aggregation;
4. *Cross-Modal Alignment*: Applies an alignment objective between utterance-level audio and textual representation spaces; and
5. *SSCL*: Adds structured supervised contrastive learning guided by the proposed continuous emotion similarity matrix.

As detailed in Table 7, incorporating adaptive layer fusion yields a substantial initial performance gain of +3.90 percentage points (78.40% to 82.30%), confirming that paralinguistic cues are distributed across intermediate transformer layers rather than concentrated in the final layer. Adding chunk weighting alone results in a minor dip to 81.40%, indicating that weighting unaligned or unregularized acoustic embeddings can introduce noise when the representation space is not yet well structured. However, pairing chunk weighting with dynamic fusion restores performance momentum to 82.90%, proving that input-dependent gating helps balance modality reliance based on utterance signal quality.

Overall, the complete DAF-CL framework achieves a net gain of +9.00 percentage points over the baseline (78.40% to 87.40%). To rigorously verify that these performance gains are statistically meaningful, we conducted a paired two-tailed t-test across five independent runs confirms that this performance improvement is statistically significant ($p < 0.001$), proving that the observed gains stem directly from our proposed cross-modal alignment and structured contrastive learning objectives rather than random variance.

4.4.2. Impact of the Chunk Aggregation Strategy

To validate our chunk aggregation design, we benchmarked five strategies for synthesizing chunk-level audio predictions into an utterance-level representation. For non-gated pooling methods (mean pooling, max pooling, and confidence-only weighting), audio and text logits were fused using fixed equal weighting (0.5), as these variants do not yield a learned dynamic gating signal.

As shown in Table 8, basic mean (83.10%) and max (82.90%) pooling perform worst due to their inability to filter out uninformative, noisy, or silent audio chunks. Weighting solely by learned acoustic saliency (α-only, 83.40%) improves performance modestly but lacks direct feedback regarding classifier certainty, making it susceptible to superficial acoustic artifacts. Conversely, confidence-only weighting (84.10%) adapts to the decision boundary but remains vulnerable to chunks on which the classifier is confidently wrong.

Combining both signals (confidence $\times$ α) achieves a performance jump to 87.40%. This dual criterion ensures that a chunk receives high weight only when it is both acoustically prominent (high α) and yields low prediction entropy (high classifier confidence), effectively penalizing deceptive segments that would otherwise mislead the final utterance-level prediction.

4.4.3. Impact of the Multimodal Fusion Strategy

We compared four distinct multimodal fusion strategies under identical evaluation settings:

1. *Early fusion*: Feature-level concatenation of audio and text embeddings followed by a joint classifier;
2. *Late fusion (Equal weighting)*: Unweighted average of independent audio and text logit predictions;
3. *Late fusion (Dedicated learnable gate)*: A separate cross-modal network that processes concatenated representations to estimate fusion weights from scratch; and
4. *Proposed Dynamic Fusion*: Reuses the chunk-level acoustic reliability score α directly as the modality weight.

As reported in Table 9, early feature concatenation yields the lowest performance (83.70%), indicating that direct concatenation without pre-alignment introduces non-linear inter-modality interference that complicates gradient optimization. Late fusion with equal weighting

Table 7: Progressive ablation study of the proposed DAF-CL framework. Each variant incrementally adds one module under identical training settings.

Model Variant	Adaptive Layer Fusion	Chunk Weighting	Dynamic Fusion	Cross-modal Alignment	SSCL	Macro F1 (%)
Baseline	–	–	–	–	–	78.40
+ Adaptive Layer Fusion	✓	–	–	–	–	82.30
+ Chunk Weighting	✓	✓	–	–	–	81.40
+ Dynamic Fusion	✓	✓	✓	–	–	82.90
+ Cross-modal Alignment	✓	✓	✓	✓	–	84.80
+ SSCL (Full DAF-CL)	✓	✓	✓	✓	✓	**87.40**

Table 8: Impact of chunk aggregation strategies on classification F_1-score.

Chunk Aggregation Method	Macro F1 (%)
Mean pooling	83.10
Max pooling	82.90
Learned α only	83.40
Confidence only	84.10
Confidence $\times$ α (Proposed)	**87.40**

Table 9: Impact of different multimodal fusion strategies on macro F_1-score.

Fusion Strategy	Macro F1 (%)
Early fusion	83.70
Late fusion (Equal weighting)	85.70
Late fusion (Dedicated learnable gate)	85.00
Proposed dynamic fusion	**87.40**

improves classification performance to 85.70%, demonstrating the stability of allowing each modality to form independent decision boundaries. However, adding a dedicated gating network reduces performance slightly to 85.00%, suggesting that unconstrained cross-modal gating networks are prone to overfitting and parameter inflation on moderate-sized emotion datasets.

In contrast, our proposed dynamic fusion strategy achieves the highest macro F_1-score (87.40%). By parameterizing the fusion weight directly with the chunk-level acoustic reliability signal α, the framework achieves adaptive, input-dependent fusion with zero parameter overhead. Reusing α acts as an effective structural regularizer, ensuring that degraded or ambiguous acoustic streams are dynamically down-weighted in favor of textual cues without introducing optimization instability.

4.4.4. Impact of the Layer Aggregation Strategy

We evaluated three approaches for aggregating the $L = 12$ hidden layer representations from the WavLM speech backbone: (1) extracting only the *last hidden layer*; (2) *uniform layer averaging*; and (3) our proposed *adaptive layer weighting* mechanism, which learns a softmax-normalized vector $\mathbf{w}$ over all hidden layers.

Table 10: Impact of layer aggregation strategy on macro F_1-score.

Layer Aggregation Method	Macro F1 (%)
Last hidden layer	82.30
Average all layers	86.50
Proposed adaptive weighting	**87.40**

Table 10 shows that extracting only the final layer yields a lower score (82.30%). Uniform layer averaging provides a significant performance increase to 86.50%, confirming that emotion-relevant paralinguistic cues are distributed across the network's depth. The proposed adaptive weighting achieves the top result (87.40%) by allowing data-driven allocation of representation importance.

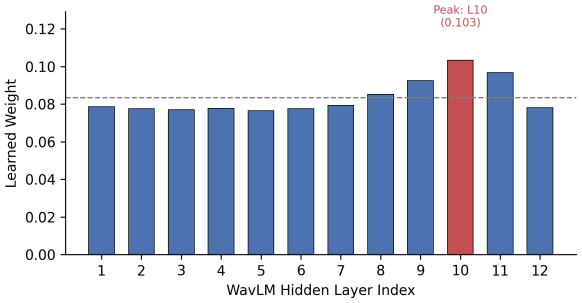


Figure 8: Learned adaptive layer weights softmax($\mathbf{w}$) across the 12 WavLM hidden layers. The dashed line marks the uniform weight ($1/L = 0.083$). Upper-middle layers (peaking at layer 10) receive higher relative emphasis, whereas early raw-acoustic layers and the final pretraining-specialized layer are down-weighted.

To visualize what the adaptive weighting mechanism learns, Figure 8 plots the softmax-normalized layer weights. The learned weights assign higher emphasis

Table 11: Impact of structured emotion similarity contrastive learning.

Contrastive Objective	Macro F1 (%)
Standard SupCon [43]	86.20
SSCL with Emotion Similarity Matrix (Proposed)	**87.40**

to upper-middle layers, peaking at layer 10 (weight = 0.103), while early layers (1–7) and the final layer (12) receive weights below the uniform baseline (0.083). This weight distribution aligns with the functional hierarchy of self-supervised speech encoders: early layers capture low-level acoustic/spectral details, upper-middle layers (8–10) encode rich prosodic and paralinguistic structure optimal for emotion discrimination, and final layers specialize toward pretraining objectives (such as pseudo-label prediction) [42]. The weights remain within a bounded range (0.074–0.103), indicating that adaptive weighting improves performance through fine-grained rebalancing rather than discarding depth.

4.4.5. Impact of the Structured Emotion Similarity Matrix

We evaluated the contribution of the proposed emotion similarity matrix by comparing Structured Supervised Contrastive Learning (SSCL) against standard Supervised Contrastive Loss (SupCon) [43], which relies on a binary treatment of class relationships by considering same-class samples as positives and different-class samples as negatives.

As reported in Table 11, replacing the proposed structured similarity formulation with conventional supervised contrastive learning reduces the Macro F1 score from 87.40% to 86.20%, corresponding to a decrease of 1.20 percentage points. This result provides empirical support for incorporating graded inter-class relationships into the contrastive objective, suggesting that preserving relative emotion similarity offers a more informative supervision signal than treating all different-class samples uniformly. While this ablation does not establish that each individual matrix value is uniquely optimal, it supports the effectiveness of introducing structured inter-emotion relationships into the learned acoustic representation space.

4.5. Sensitivity Analysis

To examine the robustness of DAF-CL with respect to its primary hyperparameters, we conduct a sensitivity analysis on the parameters governing representation learning, loss regularization, and temporal audio segmentation. Specifically, we evaluate the loss weighting coefficients λ_{align} (cross-modal alignment) and λ_{SSCL} (structured supervised contrastive learning), the temperature scaling parameter τ, as well as audio chunking parameters (chunk size c and chunk overlap ratio ρ). All sensitivity experiments were conducted under identical training and evaluation settings, with performance measured via the macro F_1-score on the unified test set.

4.5.1. Sensitivity to λ_{align}

The coefficient λ_{align} controls the relative contribution of the cross-modal alignment loss during joint optimization. This objective pulls acoustic and textual representations of the same utterance closer in the shared embedding space. An excessively small value risks insufficient cross-modal alignment, whereas an overly large value can dominate gradient updates and impair the primary classification task.

As reported in Table 12, the macro F_1-score peaks at 87.40% when $\lambda_{\text{align}} = 0.4$, which corresponds to the default setting in DAF-CL. Setting a weak weight ($\lambda_{\text{align}} = 0.1$) yields lower performance (84.42%), indicating that under-constrained alignment fails to bridge the cross-modal gap. Increasing the weight to 0.2 improves accuracy to 85.30%, confirming the benefit of aligning modality topologies. However, pushing λ_{align} to 0.6 leads to a performance drop (86.10%), demonstrating that excessive alignment regularization degrades the discriminative capacity of the classification head. Thus, $\lambda_{\text{align}} = 0.4$ provides an optimal equilibrium between cross-modal feature alignment and class discrimination.

Table 12: Sensitivity analysis of the cross-modal alignment loss coefficient λ_{align}.

λ_{align}	Macro F1 (%)
0.10	84.42
0.20	85.30
0.40 (Proposed)	**87.40**
0.60	86.10

4.5.2. Sensitivity to λ_{SSCL}

The coefficient λ_{SSCL} controls the contribution of the structured supervised contrastive learning objective. This auxiliary loss regularizes the acoustic representation space by promoting intra-class compactness and enforcing continuous topological inter-class distances via the emotion similarity matrix.

Table 13 illustrates that DAF-CL achieves its top performance (87.40%) at $\lambda_{\text{SSCL}} = 0.3$. A lower value ($\lambda_{\text{SSCL}} = 0.1$) yields 85.81%, indicating that weak contrastive guidance fails to fully enforce structured class boundaries. Increasing the weight to 0.2 improves the score to 86.15%. Conversely, an excessively

high weight ($\lambda_{\text{SSCL}} = 0.5$) slightly reduces accuracy to 86.65%, as over-emphasizing representation clustering can constrain the flexibility of the primary classifier. Across the evaluated range (0.10–0.50), performance remains consistently high (> 85.8%), demonstrating that DAF-CL is robust to variations in λ_{SSCL} while benefiting from structured contrastive regularization.

Table 13: Sensitivity analysis of the structured supervised contrastive learning coefficient λ_{SSCL}.

λ_{SSCL}	**Macro F1 (%)**
0.10	85.81
0.20	86.15
0.30 (Proposed)	**87.40**
0.50	86.65

4.5.3. Sensitivity to the Temperature Parameter τ

The temperature scaling parameter τ regulates the sharpness of the logit probability distribution within the SSCL objective. A smaller value of τ penalizes hard negative samples more severely, forcing tighter clustering, whereas a larger τ produces a smoother contrastive probability distribution that scales down gradient penalization.

As shown in Table 14, optimal performance (87.40%) is achieved at $\tau = 0.07$. Setting a smaller temperature ($\tau = 0.05$) causes a slight degradation to 86.19%, suggesting that overly aggressive hard-pair penalization can make optimization sensitive to mislabeled or ambiguous speech samples. Increasing τ to 0.10 (85.55%) and 0.20 (85.00%) leads to more pronounced performance drops, confirming that an overly smooth distribution dilutes the gradient signal necessary to push acoustically adjacent emotion categories apart. Therefore, $\tau = 0.07$ effectively balances hard-sample emphasis with stable gradient optimization.

Table 14: Sensitivity analysis of the temperature parameter τ.

τ	**Macro F1 (%)**
0.05	86.19
0.07 (Proposed)	**87.40**
0.10	85.55
0.20	85.00

4.5.4. Sensitivity to Audio Chunk Size

To process arbitrary-length speech recordings while preserving local acoustic dynamics, DAF-CL partitions audio utterances into fixed-duration chunks before aggregation. Given that the average utterance length in our dataset is approximately 3 s, we evaluated chunk durations $c \in \{2, 3, 4\}$ s to determine the optimal trade-off between temporal resolution and contextual coverage.

As reported in Table 15, the macro F_1-score exhibits high stability across variations in chunk duration. A duration of $c = 3$ s achieves the highest performance (87.40%). Shorter 2 s chunks yield 86.27%, as limited window length can fragment sustained prosodic contours. Longer 4 s chunks achieve 86.68%, where broader context comes at the cost of reduced temporal granularity for localized emotional bursts. A 3 s window matches the natural acoustic phrase structure of emotional speech, establishing an effective balance between local resolution and global context.

Table 15: Sensitivity analysis of the audio chunk size c.

Chunk Size (s)	**Macro F1 (%)**
2	86.27
3 (Proposed)	**87.40**
4	86.68

4.5.5. Sensitivity to Chunk Overlap Ratio

To prevent boundary artifacts and ensure continuous feature extraction across adjacent chunks, audio segmentation is performed with a temporal overlap ratio ρ. Higher overlap preserves transitional prosodic cues across segment boundaries but increases computational complexity and feature redundancy.

Table 16 shows the performance impact of varying the overlap ratio $\rho \in \{0.25, 0.50, 0.75\}$. A low overlap ratio ($\rho = 0.25$) results in a drop to 85.10%, indicating that minimal overlap loses crucial acoustic information occurring at chunk boundaries. Increasing the overlap to $\rho = 0.50$ significantly improves accuracy to 87.40%. Further increasing the overlap to $\rho = 0.75$ yields a comparable score of 87.00% while incurring a 50% increase in forward-pass compute cost due to extra chunk processing. Consequently, $\rho = 0.50$ is selected as it achieves optimal performance while avoiding unnecessary computational overhead.

Table 16: Sensitivity analysis of the audio chunk overlap ratio ρ.

Chunk Overlap (ρ)	**Macro F1 (%)**
0.25	85.10
0.50 (Proposed)	**87.40**
0.75	87.00

4.6. Representation Analysis

To qualitatively investigate the effect of the proposed structured supervised contrastive learning (SSCL) objective on the learned audio representation space, we apply t-SNE [44] to the chunk-level audio embeddings obtained before and after incorporating SSCL. A balanced subset of 1,000 embeddings is randomly sampled from the test set, comprising 250 instances per emotion class. The identical sample subset and t-SNE hyperparameters are maintained across both configurations to ensure a consistent visual comparison.

Figure 9 illustrates the resulting two-dimensional embedding distributions. Prior to incorporating SSCL (Figure 9a), the audio embeddings exhibit substantial inter-class overlap and poorly defined boundaries, indicating a low degree of discriminative structure in the acoustic space. After integrating the SSCL objective (Figure 9b), the representation space demonstrates significantly enhanced intra-class compactness and superior inter-class separation.

Despite the overall structural improvement, moderate manifold overlap persists, particularly along the boundary between the *Happy* and *Neutral* emotion categories. This visual behavior aligns directly with the quantitative confusion matrices in Section 4.3, where acoustic similarities between high-arousal positive speech and neutral tone lead to minor inter-class confusion.

These visual shifts provide clear qualitative support for the functional role of SSCL in shaping the latent space. By parameterizing the contrastive objective with the continuous emotion similarity matrix, SSCL enforces discriminative cluster margins while preserving soft topological affinities among prosodically adjacent emotions. Because t-SNE is a non-linear projection technique that can introduce local spatial distortions, this visualization serves as complementary qualitative validation rather than a standalone metric of representation quality.

5. Limitations and Future Work

Despite its robustness across different testing conditions and its dynamic modality weighting, DAF-CL still has areas that require further investigation. Although the external zero-shot results provide evidence of transferability, they do not establish robustness across different languages, domains, and recording conditions. Future work should extend the evaluation to additional languages and conditions to better assess the model's generalization ability.

Since DAF-CL combines acoustic and textual information, errors in ASR-generated transcripts may affect modality weighting, causing the model to rely on unreliable textual information. Future work could evaluate the model under controlled ASR error rates, compare reference and generated transcripts, and investigate uncertainty-aware fusion to reduce reliance on unreliable text.

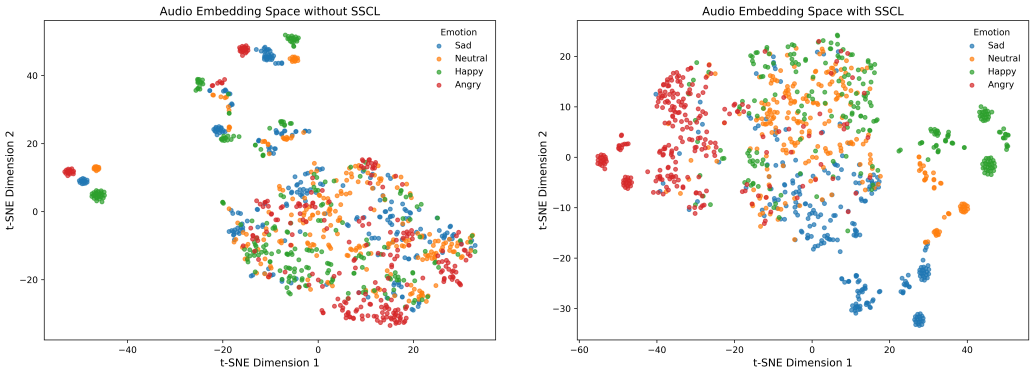
The model's pipeline includes adaptive layer aggregation, overlapping audio chunks, a text encoder, cross-modal alignment during training, and ensemble inference, which may increase computational costs. Future work should measure inference latency, memory consumption, throughput, and energy usage, and investigate chunk reduction, encoder distillation, quantization, and more efficient fusion.

Although structured supervised contrastive learning incorporates emotion-related similarities, some class boundaries remain difficult to distinguish, particularly between happy and neutral expressions. Future work could investigate uncertainty-aware prediction to better handle ambiguous emotional expressions.

Addressing these limitations could improve the robustness and efficiency of DAF-CL and support its practical deployment in emotion recognition applications, including emotion-aware large language models (LLMs).

6. Conclusions

In this paper, we introduced DAF-CL, a novel dynamic multimodal fusion framework designed for robust speech and text emotion recognition. To address the inherent challenges of modality reliability and representation heterogeneity, the framework integrates a confidence-aware weighting mechanism with a comprehensive multi-objective optimization strategy. Specifically, DAF-CL leverages adaptive layer aggregation for acoustic feature extraction, cross-modal alignment objectives, and structured supervised contrastive learning ($\mathcal{L}_{\text{SSCL}}$) to enforce distinct emotional separation in the shared embedding space. Extensive empirical evaluations demonstrate that DAF-CL achieves substantial performance gains—advancing from a baseline accuracy of 78.40% to a final accuracy of 87.40% ($p < 0.001$) under rigorous speaker-independent partitioning and 5-fold cross-validation. Furthermore, zero-shot evaluations on unseen external datasets confirm the model's strong cross-domain transferability and generalization capacity. Comprehensive ablation and sensitivity analyses validate the underlying architectural choices, confirming that dynamic reliability assessment and multi-objective guidance are essential for optimal multimodal integration. Ultimately, DAF-CL provides a principled,



(a) Without SSCL

(b) With SSCL

Figure 9: t-SNE visualization of the chunk-level audio embedding manifold (a) without and (b) with the proposed SSCL objective. Both plots display the exact same balanced subset of 1,000 embeddings (250 instances per emotion class).

high-performance architecture for affective computing and advanced information fusion.

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